Advanced Applications in Agricultural Systems Modelling

Module 6 - ABE 721: Agricultural Systems Modelling and Simulation

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Lecture Outline

1. 1. Comprehensive Crop Simulation
2. 2. Spatial Modelling and GIS Integration
3. 3. Soft Computing: Artificial Neural Networks
4. 4. Computer-Aided Manufacturing (CAM)
5. 5. Bio-Economic Optimization & DSS

Advanced Crop Simulation Models

Previous modules utilized simplified logistic equations. Real-world engineering requires comprehensive suites like **DSSAT** or **APSIM**.

- Consist of dozens of tightly coupled mechanistic sub-models.
- Sub-models include: Phenology, Carbon Assimilation, Soil Hydrology, and Nitrogen Cycling.
- Requires transition from chronological time to biological time.

The Mathematics of Phenology

Phenology is the study of periodic biological life cycle events.

In mechanistic models, chronological time (days) is functionally useless (a plant grows faster at 30°C than 15°C). We use **Thermal Time** or **Growing Degree Days (GDD)**.

Basic Thermal Accumulation

$$\Delta TT = \max\left(0, \frac{T_{max} + T_{min}}{2} - T_{base}\right)$$

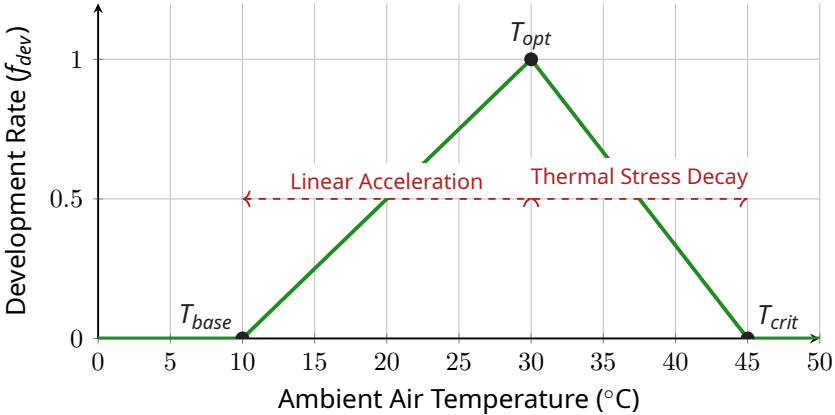
Piecewise Thermal Time Formulation

Linear approximations fail during severe heatwaves. Advanced models use a truncated continuous integral over the diurnal temperature curve $T(t)$:

$$\Delta TT = \int_{0}^{24} f_{dev}(T(t))\, dt$$

Where the biological response function limits growth past T_{opt} and halts it completely at T_{crit}.

Phenological Thermal Response Curve



Biomass Assimilation

Daily gross photosynthesis (P_{gross}) is determined by intercepted Photosynthetically Active Radiation (PAR) and Radiation Use Efficiency (RUE):

$$P_{gross} = RUE \times PAR \times (1 - \exp(-k \cdot LAI))$$

Net daily accumulation of biomass (W) requires subtracting maintenance respiration (R_m):

$$\frac{dW}{dt} = Y_g \left(P_{gross} - R_m \cdot W\right)$$

The Partitioning Matrix

New biomass (ΔW) must be dynamically distributed to different anatomical organs using a partitioning vector $\alpha(t)$:

$$\alpha(t) = \begin{bmatrix} \alpha_{root}(t) \\ \alpha_{stem}(t) \\ \alpha_{leaf}(t) \\ \alpha_{grain}(t) \end{bmatrix}, \quad \text{subject to} \quad \sum \alpha_i(t) = 1$$

- Vegetative stage: $\alpha_{grain} = 0$.
- Post-anthesis (flowering): $\alpha_{root/leaf} \to 0$, and $\alpha_{grain} > 0.8$.

Spatial Modelling and GIS

Lumped (0D) models fail to capture the topographical, hydrological, and agronomic variability of large-scale farming.

- Precision agriculture explicitly requires Geographic Information Systems (GIS).
- Continuous geographic reality must be discretized into mathematical data structures.
- Two primary architectures: **Vector** and **Raster**.

1. Vector Data Architecture

Vector systems use discrete geometric primitives (points, lines, polygons) defined by Cartesian coordinate matrices.

$$\mathbf{P} = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_n & y_n \\ x_1 & y_1 \end{bmatrix}$$

Calculating exact field area using the **Shoelace Formula**:

$$Area = \frac{1}{2} \left| \sum_{i=1}^{n-1} (x_i y_{i+1} - x_{i+1} y_i) \right|$$

2. Raster Data Architecture

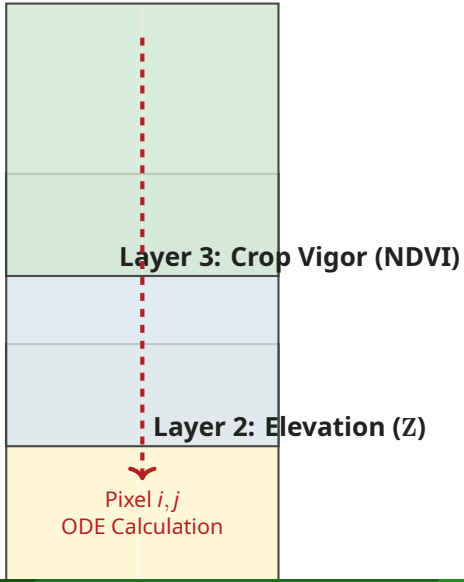
Raster systems discretize space into a regular 2D matrix of cells (pixels).

Map Algebra

Performing localized mathematical operations across multiple spatially aligned matrices. E.g., Calculating Crop Stress ($\mathbf{S}$) from Moisture ($\mathbf{M}$) and Demand ($\mathbf{D}$):

$$\mathbf{S} = \mathbf{M} - \mathbf{D} \implies S_{i,j} = M_{i,j} - D_{i,j} \quad \forall i, j$$

3D Map Algebra Integration



Distributed Modelling: The USLE

The **Universal Soil Loss Equation (USLE)** models watershed soil erosion using distributed raster map algebra:

USLE Formulation

$$A(x,y) = R(x,y) \cdot K(x,y) \cdot LS(x,y) \cdot C(x,y) \cdot P(x,y)$$

- R: Rainfall Erosivity (Radar).
- K: Soil Erodibility.
- LS: Topographic Length-Slope (DEM Differential Geometry).
- C: Cover Management (Satellite NDVI).

Spatial Interpolation: Kriging

Sensor data (e.g., soil nitrate probes) are expensive and collected at sparse point locations. We must interpolate to create continuous raster maps.

Kriging minimizes the variance of the estimation error using a probabilistic variogram. The estimated value $\hat{Z}$ at location $\mathbf{s}_0$ is:

$$\hat{Z}(\mathbf{s}_0) = \sum_{i=1}^{n} \lambda_i Z(\mathbf{s}_i)$$

The Mathematics of Kriging

The optimal weights (λ_i) are found by solving a linear system based on spatial covariance (γ), ensuring an unbiased estimator ($\sum \lambda_i = 1$):

$$\begin{bmatrix} \gamma_{11} & \cdots & \gamma_{1n} & 1 \\ \vdots & \ddots & \vdots & \vdots \\ \gamma_{n1} & \cdots & \gamma_{nn} & 1 \\ 1 & \cdots & 1 & 0 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_n \\ \mu \end{bmatrix} = \begin{bmatrix} \gamma_{10} \\ \vdots \\ \gamma_{n0} \\ 1 \end{bmatrix}$$

Why Soft Computing?

Chaotic agricultural systems frequently defy strict mechanistic PDE modelling. Parameters may be completely unmeasurable.

- We deploy **Artificial Neural Networks (ANNs)**.
- A massive paradigm shift from physics-based modelling to pure data-driven machine learning.

Multi-Layer Perceptron (MLP) Architecture

The standard ANN architecture for predicting yields or streamflow.

Three Layer Types:

1. **Input Layer:** Raw exogenous variables (Temp, Humidity).
2. **Hidden Layers:** Deep computational nodes executing non-linear transformations.
3. **Output Layer:** Generates the final predicted system state ($\hat{y}$).

Feedforward Mathematics

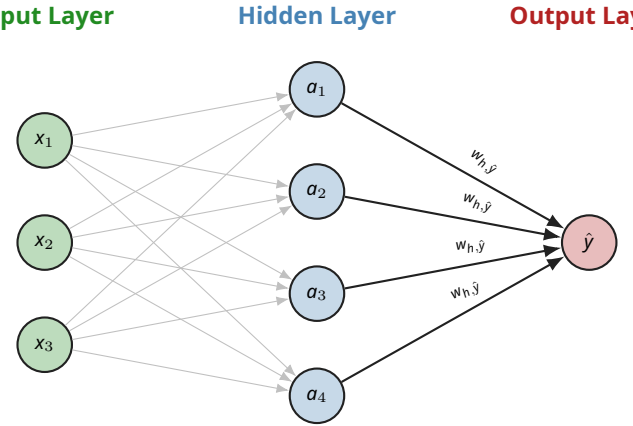
The linear summation arriving at a neuron j is the dot product of the inputs and weights, plus a bias:

$$z_j = \sum_{i=1}^{n} w_{ij} x_i + b_j = \mathbf{w}_j^T \mathbf{x} + b_j$$

This is passed through a non-linear **Activation Function** $f(z)$ (e.g., Sigmoid) to grant the network non-linear modelling power:

$$a_j = \sigma(z_j) = \frac{1}{1 + e^{-z_j}}$$

Neural Network Topology



Training the Network: Backpropagation

We must ”train” the random matrix weights using historical data. This is a massive optimization problem minimizing the Cost Function J:

$$J(\mathbf{W}, \mathbf{b}) = \frac{1}{2m} \sum_{k=1}^{m} \left(\hat{y}^{(k)} - y_{true}^{(k)} \right)^2$$

Calculating the error gradient of a deep hidden weight requires the calculus **Chain Rule** (Backpropagation).

Calculus of Backpropagation

For a specific weight $w_{ij}^{(l)}$, the gradient is calculated backwards from the final output error:

$$\frac{\partial J}{\partial w_{ij}^{(l)}} = \frac{\partial J}{\partial a_j^{(l)}} \times \frac{\partial a_j^{(l)}}{\partial z_j^{(l)}} \times \frac{\partial z_j^{(l)}}{\partial w_{ij}^{(l)}}$$

Simplifying this using the recursive error term ($\delta_j^{(l)}$):

$$\frac{\partial J}{\partial w_{ij}^{(l)}} = \delta_j^{(l)} \cdot a_i^{(l-1)}$$

Gradient Descent Weight Update

Once all gradients are computed using backpropagation, the computer updates the entire network simultaneously:

Weight Update Equation

$$w_{ij}^{(new)} = w_{ij}^{(old)} - \alpha \frac{\partial J}{\partial w_{ij}}$$

- α represents the **Learning Rate**.
- Repeating this millions of times allows the ANN to ”learn” the non-linear correlations of the agricultural system.

The CAD-to-CAM Pipeline

A digital CAD model (Module 1) cannot plow a field. It must be translated into physical steel using Computer Numerical Control (CNC).

1. **CAD Modelling:** Boundary Representation (B-Rep) solid model.
2. **CAM Toolpath Generation:** Software calculates the optimal cutting paths.
3. **Post-Processing:** Translating the path into **G-Code** (RS-274) coordinates.

CNC Kinematics

Executing G-code requires intense mathematical kinematics.

Modern 5-axis CNC machines (for complex agricultural turbine impellers) handle kinematics using **Homogeneous Transformation Matrices** (4×4).

- Combines spatial translations (T_x, T_y, T_z) and rotations ($\theta_x, \theta_y, \theta_z$) into a single matrix multiplication operation.

Homogeneous Transformation Matrices

The transformation from the machine base to the tool tip:

$$\mathbf{T} = \begin{bmatrix} R_{11} & R_{12} & R_{13} & T_x \\ R_{21} & R_{22} & R_{23} & T_y \\ R_{31} & R_{32} & R_{33} & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example: Pure Z-axis rotation (θ):

$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Whole-Farm Mathematical Programming

The normative purpose of modelling is to **optimize**. We use physical simulation outputs (yields) as inputs for economic optimization algorithms (Decision Support Systems).

Linear Programming (LP) maximizes profit allocating constrained resources (land, labor, water) across competing crops.

Standard LP Formulation

Let x_j be hectares of crop j, and c_j be net profit per hectare. Let b_i be available resources, and a_{ij} be the technical requirement.

Objective Function (Maximize Profit Z):

$$\text{Maximize } Z = \sum_{j=1}^{n} c_j x_j$$

Subject to Constraints (Resource limits):

$$\sum_{j=1}^{n} a_{ij} x_j \leq b_i \quad \text{and} \quad x_j \geq 0$$

LP Matrix Formulation

$$\begin{aligned} \text{Maximize} \quad & Z = \mathbf{c}^T \mathbf{x} \\ \text{Subject to} \quad & \mathbf{A}\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

- Constraints form a multi-dimensional hyper-polygon.
- The **Simplex Algorithm** systematically pivots along the outer edges to find the absolute highest profit vertex.

Shadow Prices and Dual Optimization

Solving the LP also yields the **Shadow Price** (λ_i) for every constrained resource.

$$\lambda_i = \frac{\partial Z^*}{\partial b_i}$$

- λ_i represents the exact marginal increase in total profit if the farmer acquired exactly one more unit of resource *i*.
- *DSS Logic:* If the shadow price of water is 500 TZS/m^3, it is profitable to buy water from a neighbor *only if* they charge less than 500 TZS.

Summary of Module 6

- **Crop Models:** Combine phenology (Thermal Time) with mass-balance biomass partitioning matrices.
- **GIS Integration:** Distributes models over vectors and rasters using Map Algebra and spatial interpolation (Kriging).
- **Soft Computing:** ANNs learn non-linear physics via the Backpropagation algorithm.
- **CAM & LP:** We move from digital CAD to physical CNC kinematics, and optimize the final farm operations using Linear Programming (Simplex).

End of Module 6

Questions & Discussion